

8872/01

19 September 2013

50 minutes

READ THESE INSTRUCTIONS FIRST

Write your name, class and shade your exam index number on the Answer Sheet in the spaces provided.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

A *Data Booklet* is provided. Do not write anything on the *Data Booklet*.

SECTION A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

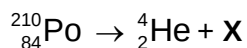
1. The relative abundances of the isotopes of a sample of titanium are shown in the table below.

Relative Isotopic Mass	46	47	48	49	50
Relative Abundance	11.2	10.1	100.0	7.3	7.0

What is the relative atomic mass of titanium in this sample?

- A** 47.92
B 47.91
C 47.90
D 47.89
2. How many atoms are there in 500 cm³ of oxygen at 0 °C and 1 atm?
- A** 1.25×10^{22}
B 1.34×10^{22}
C 2.50×10^{22}
D 2.68×10^{22}
3. The radioactive isotope $^{210}_{84}\text{Po}$ was said to be the agent that poisoned the former Russian security agent Alexander Litvinenko in London in November 2006.

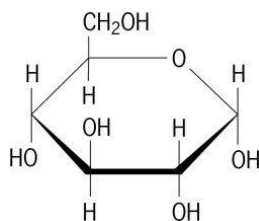
$^{210}_{84}\text{Po}$ decays to give an element **X** and emits a high energy α -particle (which is a helium nucleus, ^4_2He). No other particle is produced. α -particles cause irreparable damage to the tissues of internal organs.



Which row in the table correctly describes the composition of $^{210}_{84}\text{Po}$ and element **X**?

	$^{210}_{84}\text{Po}$	X	
	number of neutrons	number of protons	number of neutrons
A	126	80	122
B	126	82	124
C	210	80	206
D	210	82	208

4. In the body, cellular respiration produces energy from the oxidation of glucose. The diagram shows the structure of glucose.



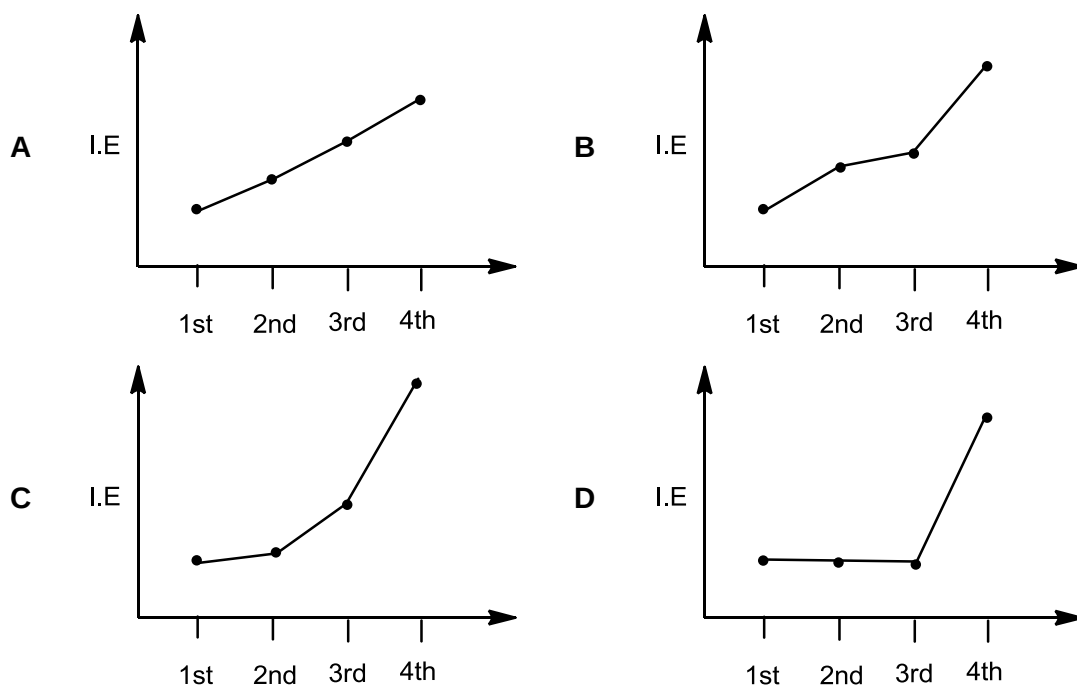
A new artificial sweetener has been produced by replacing all of the hydroxyl groups attached directly to the ring carbon atoms in glucose with chlorine atoms.

What is the empirical formula of this chlorinated glucose?

- | | |
|--|---|
| <p>A CHClO</p> <p>B $\text{C}_3\text{H}_4\text{Cl}_2\text{O}$</p> | <p>C $\text{C}_6\text{H}_7\text{Cl}_5\text{O}$</p> <p>D $\text{C}_6\text{H}_8\text{Cl}_4\text{O}_2$</p> |
|--|---|
5. What is the electronic configuration of an element with **first** ionisation energy higher than that of each of its neighbours in the Periodic Table?
- A** $1s^2 2s^2 2p^6 3s^2 3p^1$
- B** $1s^2 2s^2 2p^6 3s^2 3p^2$
- C** $1s^2 2s^2 2p^6 3s^2 3p^3$
- D** $1s^2 2s^2 2p^6 3s^2 3p^5$
6. Which one of the following does **not** contain any unpaired s or p electron?

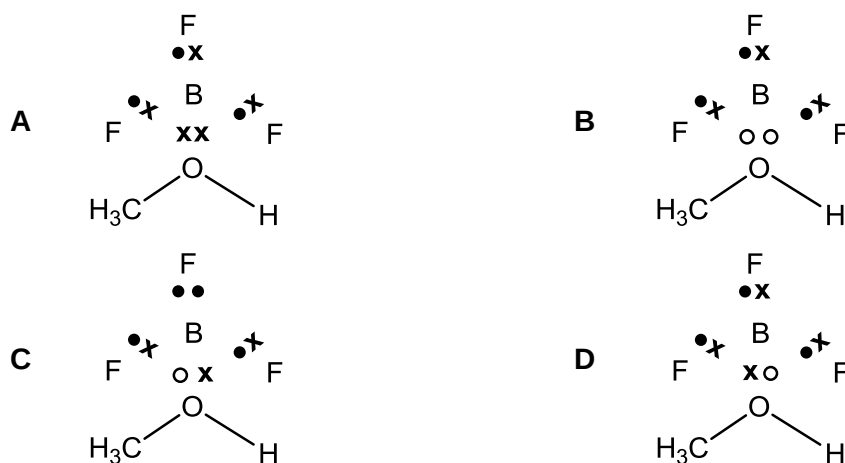
- | | |
|--|---|
| <p>A Cr</p> <p>C S^-</p> | <p>B Ge</p> <p>D Sc</p> |
|--|---|

7. Which graph correctly shows the first four ionisation energies of a Group III element?

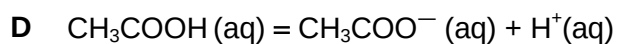
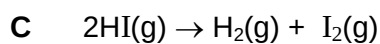
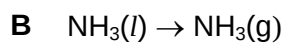
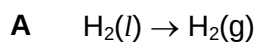


8. $\text{BF}_3 \cdot \text{CH}_3\text{OH}$ is used to produce methyl esters from compounds containing acyl groups. In the diagrams, $\times$, $\bullet$ and $\circ$ represent electrons from B, F and O respectively.

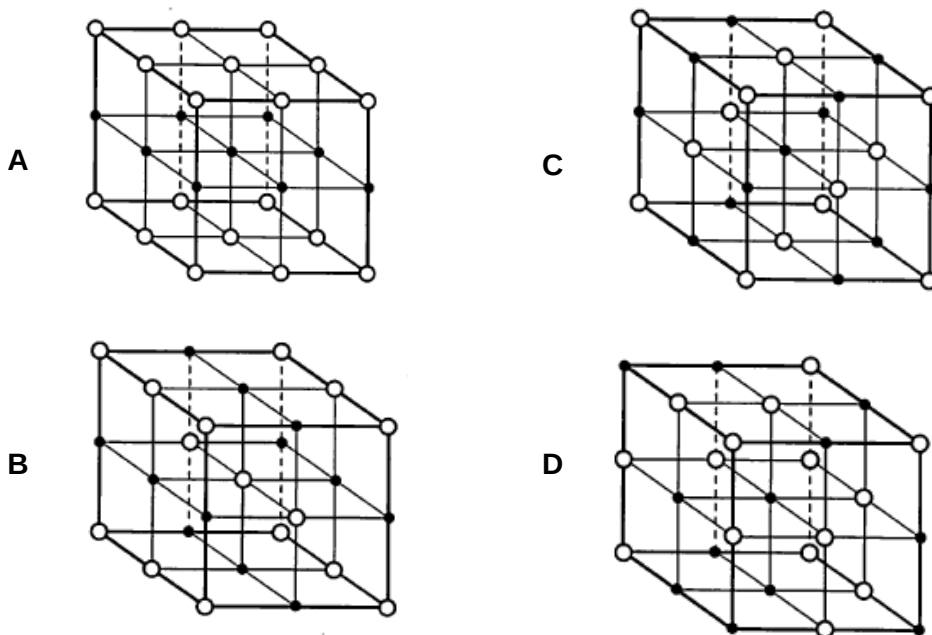
Which tetrahedral structure correctly shows the electron pairs around the boron atom?



9. Which process involves the breaking of hydrogen bonds?



10. Which diagram best represents the structure of solid magnesium oxide?



11. Which of the following is involved in determining the enthalpy change of a chemical reaction?

- A the number of stages involved in the chemical reaction
- B the activation energy of the reaction
- C the initial and final state of the reacting system
- D the intermediates of the overall chemical reaction

12. Which class of reaction always has an endothermic enthalpy change?

- | | |
|--------------|------------------|
| A combustion | C formation |
| B ionisation | D lattice energy |

13. If the reaction $\text{P} + \text{Q} \rightarrow \text{R} + \text{S}$ is described as being *zero order* with respect to **P**, it means that

- A **P** is a catalyst in this reaction.
- B no **P** molecules possess sufficient energy to react.
- C the concentration of **P** does not change during the reaction.
- D the rate of reaction is independent of the concentration of **P**.

14. The reaction of a compound **RS** is shown below.

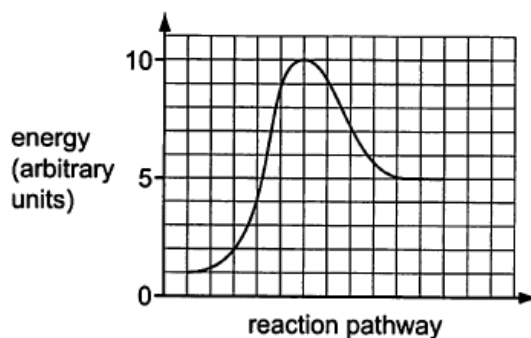


The rate equation for the reaction is $\text{rate} = k[\text{RS}]$ and the rate constant is found to be $3.6 \times 10^{-3} \text{ s}^{-1}$.

If the initial concentration of **RS** is $2.0 \times 10^{-2} \text{ mol dm}^{-3}$, what will be the concentration of **RS** and **R** after 385 seconds?

	[RS] / mol dm^{-3}	[R] / mol dm^{-3}
A	2.5×10^{-3}	2.5×10^{-3}
B	2.5×10^{-3}	1.75×10^{-2}
C	5.0×10^{-3}	5.0×10^{-3}
D	5.0×10^{-3}	1.5×10^{-2}

15. The diagram shows the reaction pathway diagram for an uncatalysed reaction.



The reaction is then catalysed.

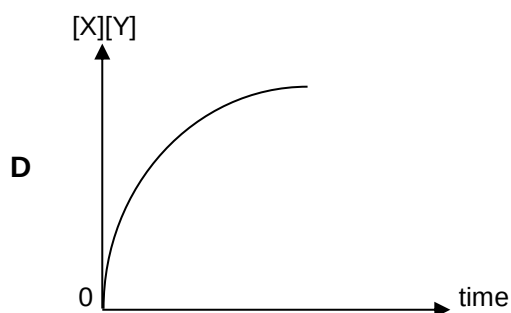
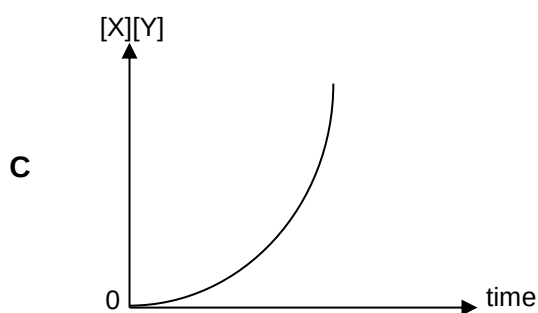
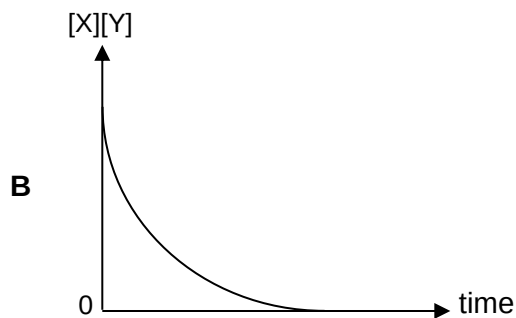
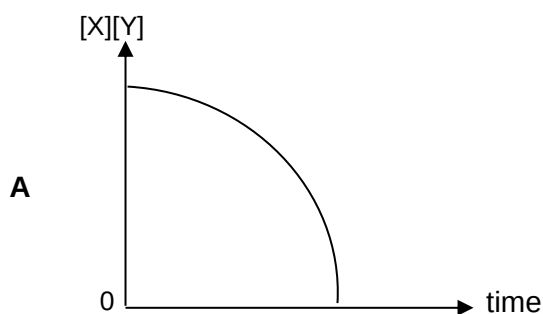
What are the changes in the rate constant and the reaction pathway diagram?

	rate constant	energy profile
A	decrease	
B	decrease	
C	increase	
D	increase	

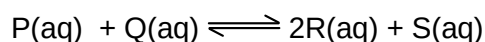
16. The product of the concentrations of X and Y, $[X][Y]$, is plotted against time, t , for the following second-order reaction.



Which graph would be obtained?



17. An equilibrium can be represented by the following equation.



In a certain mixture, the equilibrium concentration of **S** is 10 mol dm^{-3} .

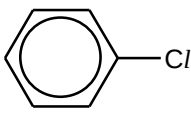
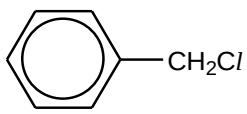
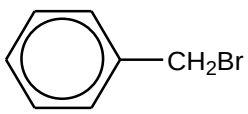
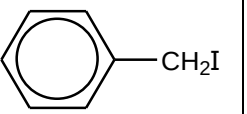
What will be the new equilibrium concentration of **S** if 5 mol of **Q** is dissolved in the mixture?

- A** 15 mol dm^{-3}
 - B** Between 10 mol dm^{-3} and 15 mol dm^{-3}
 - C** 10 mol dm^{-3}
 - D** Between 5 mol dm^{-3} and 10 mol dm^{-3}
18. Which of the following is an **incorrect** trend of the elements of the Period 3 of the Periodic Table?
- A** The radii of the atoms decrease.
 - B** The oxides of the elements change from basic to acidic.
 - C** The melting points of the elements decrease steadily.
 - D** The compounds of the elements change from ionic to covalent.

19. The oxide of **G** and the chloride of **H** are separately mixed with water. The two resulting mixtures have the same effect on litmus. Which of the following represents the correct elements?

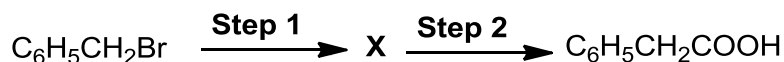
	G	H
A	Al	Si
B	Mg	P
C	P	Al
D	S	Na

20. The table below shows the rate of hydrolysis of halogen-containing compounds **P** to **S**. The rate of hydrolysis is measured by the speed at which the silver halide precipitate is formed.

P	Q	R	S
			
No precipitate.	Precipitate formed after 10 minutes.	Precipitate formed after 2 minutes.	Precipitate forms instantaneously.

Which statement explains the rate of reaction?

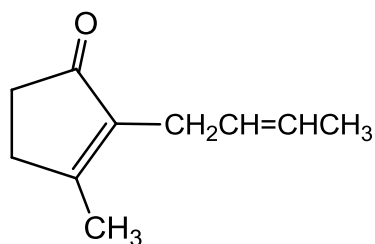
- A** The C–X bond length increases from compound **P** to **S**.
B The solubility of the compounds increases from **P** to **S**.
C The mechanism for the reaction changes from **P** to **S**.
D The reaction conditions are increasingly milder from **P** to **S**.
21. Phenylethanoic acid, $\text{C}_6\text{H}_5\text{CH}_2\text{COOH}$ can be synthesised from (bromomethyl)benzene, $\text{C}_6\text{H}_5\text{CH}_2\text{Br}$ in the laboratory in two steps.



What are reagents and conditions that can be used for this synthesis?

	Step 1	Step 2
A	NaOH(aq), heat	Acidified $\text{K}_2\text{Cr}_2\text{O}_7$, heat
B	Ethanollic NaOH, heat	Acidified KMnO_4 , heat
C	Ethanollic NaCN, heat	HCl(aq), heat
D	Cold HCN, trace amount of NaCN	H_2SO_4 (aq), heat

22. Jasmone is an active ingredient of jasmine. It is extracted from jasmine flowers for perfume.

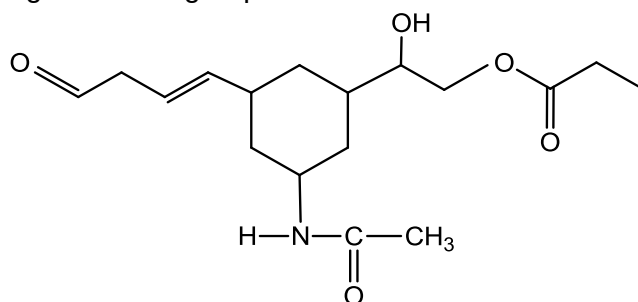


Jasmone

Which of the following statements is correct?

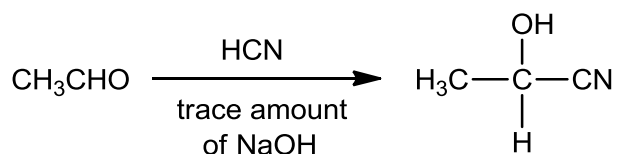
- A** It exists as 1 pair of *cis-trans* isomers.
- B** There are 4 sp^3 carbon atoms in Jasmone.
- C** Effervescence is observed when Jasmone is heated with acidified $KMnO_4$.
- D** A reddish brown precipitate is observed when hot alkaline Cu^{2+} complex solution was added to Jasmone.
23. Which transformation can be readily achieved by only one substitution reaction?
- A** $CH_3CH=CH_2 \rightarrow CH_3CH=CHCl$
- B** $(CH_3)_3CCH_2F \rightarrow (CH_3)_3CCH_2NH_2$
- C**
- D**

24. Which of the following functional group is found in the molecule below?



- A** Tertiary alcohol
- B** Amide
- C** Ketone
- D** Phenyl group

25. Ethanal reacts with HCN as shown below.



When the reaction is repeated with propanone, the reaction is slower.

Which of the following helps to explain the above observation?

- A C=O in propanone is stronger.
- B Propanone molecules are held by stronger permanent-dipole permanent-dipole interaction.
- C $-\text{CH}_3$ is electron-donating, it intensifies the negative charge in propanone.
- D $-\text{CH}_3$ is electron-donating, it makes the carbonyl carbon in propanone less electron-deficient.

SECTION B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

26. An ionic compound **Q**

- Has an empirical formula NH_2O .
- Reacts with NaOH(aq) to produce ammonia gas

Which statements about the anion in this salt are correct?

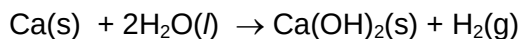
- 1 It has a single negative charge
- 2 It contains an atom with an oxidation number of +3.
- 3 It contains 20 electrons.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 27.** Calcium reacts with water to form calcium hydroxide and hydrogen.



The standard enthalpy change for this reaction can be measured in the laboratory.

What further information is needed in order to calculate the standard enthalpy change of formation of calcium hydroxide, $\Delta H_f^\circ(\text{Ca(OH)}_2)$?

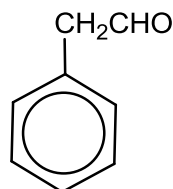
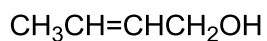
- 1** ΔH_f° of $\text{H}_2\text{O(l)}$
- 2** ΔH_f° for $\text{H}_2\text{(g)}$
- 3** First and second ionisation energies of Ca

- 28.** A catalytic converter is part of the exhaust system of many modern cars.

Which reactions occur in a catalytic converter?

- 1** $2\text{CO} + 2\text{NO} \rightarrow 2\text{CO}_2 + \text{N}_2$
- 2** $\text{CO}_2 + \text{NO} \rightarrow \text{CO} + \text{NO}_2$
- 3** $2\text{SO}_2 + 2\text{NO} \rightarrow 2\text{SO}_3 + \text{N}_2$

- 29.** Which of the following tests can be used to distinguish between the two organic compounds below?



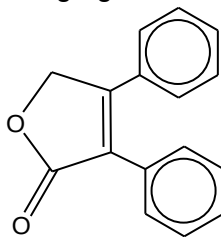
- 1** Na metal
- 2** Aqueous bromine, in the dark
- 3** 2,4-Dinitrophenylhydrazine

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

30. A derivative of *Rofecoxib*, an efficient drug against arthritis, has the following structure.



Which of the following statements about this compound are correct?

- 1** It undergoes hydrogenation with hydrogen gas and nickel catalyst.
- 2** It reacts with cold dilute alkaline KMnO_4 to give a diol.
- 3** The molecule is planar.